

Adroit Technologies

Protocol Driver

Name	Siemens Ethernet 2nd Generation (ISO over TCP)
DLL	SIEMENS



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1. Overview

The Adroit Siemens protocol driver supports a set of various PLC models and communication protocols.

1.1. Supported Communication Protocols

- **ISO over TCP/IP** – Ethernet connection.
 - supported PLC models: S7-200, S7-300 and S7-400 and S7-1200 and S7-1500
- **MPI** – Serial connection.
 - supported PLC models: S7-300 and S7-400 and S7-1200 and S7-1500
- **MPI via IBH Link S7/S7++** - Ethernet connection from Adroit to IBH Link, serial connection from IBH Link to PLC.
 - supported PLC models: S7-300 and S7-400 and S7-1200 and S7-1500
- **PPI** - Serial connection.
 - supported PLC models: S7-200
- **PPI via IBH Link S7/S7++** - Ethernet connection from Adroit to IBH Link, serial connection from IBH Link to PLC.
 - supported PLC models: S7-200

See [Appendix A](#) for a list of tested CPU's and CP's and their respective part numbers.

See [Appendix B](#) for some considerations about the protocol and how it is used effectively.

See [Appendix C](#) for MicroWin S7-200 configuration.

See [Appendix D](#) for Siemens TIA Portal configuration for S7-1200/S7-1500.

The system requires no third party drivers, software, stacks or plc code to run, provided the Siemens PLC and/or IBH Link and/or CP card is supported and has been configured appropriately.

2. Revisions

2.1. Document Revision

Rev.	Date	Author	Comment	Approval
1.0	14-Dec-2022	J. Duncan	New Format	D. Jordaan
2.0	06-Jan-2022	K. Robinson	Grammar and proofing	D. Jordaan
3.0				
4.0				
5.0				
6.0				
7.0				

2.2. DLL Revision

Rev.	Date	Changes
1	30-Oct-2013	First release of the driver.
2	15-Mar-2016	Added S7-1200 and S7-1500 TIA Configuration.
31	19-Oct-2020	Added Reset and Switch-Connection support to driver from Device Agent.
31	20-Oct-2020	Added Communication type to driver options (PG/OP/S7)
32	11-Nov-2020	Added PLC side watchdog mechanism

3. Wiring

Please see standard Ethernet wiring documentation for ISO-TPC, PPI-IBH and MPI-IBH protocols.

4. Device Configuration

4.1. Device Configuration

Siemens : S1

Reduncy

☒ Enable Secondary

☒ Primary
☐ Secondary

Device Details

PLC Model	S7 300
Protocol	ISOTCP
Retry count	3
Backoff Period (ms)	0
Connection Timeout (ms)	1000
PDU Length (bytes) (4->222)	200

OK

Cancel

Advanced

Help

Connection Configuration

Ethernet Configuration

IP Address: 10 . 10 . 10 . 215

Port: 102

Ping Test

Serial Configuration

Com Port:

Baud Rate: 9600

Parity: Even

Protocol Configuration

Rack: 0

Slot: 2

Type: S7(3)

MPI Configuration

Local Address: 0

PLC Address: 2

Speed: 187k

Test Device Configuration for Connectivity

Test Address: DB1.DBW0

ID Type: Integer

Start Reading

Watchdogs

☐ Enable PLC program running check

Address: DB: 0 DBW: 0

☒ Enable driver send watchdog

Bit address: MX100.0

Timeout (mSec): 3000

Type: PulseON

Using the LibNoDave communication library, by Thomas Hergenahn

4.1.1. Redundancy

Enable Secondary – (Default = unchecked) Turn this option to have the driver maintain 2 channels of communication, where the radio buttons **Primary** and **Secondary** can then be used to toggle between configurations.

4.1.2. Device Details

PLC Model – (Default = S7-200) Select which PLC model the driver will connect to, S7 200, S7 300 or S7 400. In the case of using S7-1200 or S7-500 select the S7300 option.

Protocol – Select which communication protocol should be used when the driver communicate with the PLC. See the [Overview: Supported Communication Protocols](#) section for more details on supported protocols.

Retry count – In the event of communication failure, this value indicates the number of retry attempts before the driver enters a back-off state.

Back-off Period - The time in milliseconds which the driver idles when in the back-off state.

Connection Timeout – Maximum time in milliseconds Adroit will wait for a device to reply, before timing out and retrying.

PDU Length - The maximum bytes to read in a single transaction.

4.1.3. Connection Configuration

Ethernet Configuration

This block will only be active if either of the following protocols are selected: ISOTCP, MPI by IBH, or PPI by IBH.

IP Address - Configurable Ethernet Address. 127.0.0.1 (default).

Port – Usually 102 when ISOTCP is selected and 1099 when MPI by IBH, or PPI by IBH is selected. Change not recommended.

Serial Configuration

This block will only be active if either of the following protocols are selected: PPI, MPI

Com Port – The available serial ports on your machine.

Baud Rate – The speed of data transmission. Any of the following Baud Rates may be specified: 300, 600, 1200, 2400, 4800, 9600, 19200, 38400, 57600, 115200.

Parity – The parity bit setting used in the data transmission. This can be Even, Odd or None.

4.1.4. Protocol Configuration

Rack – The rack number in which the PLC is installed. The rack is only applicable to the S7-300 and S7-400 PLCs. All the other PLC models have no concept of racks and is therefore always 0.

Slot – The slot number in which the CPU is installed. By default if using the S7-300 the PSU takes slot 1 and the CPU slot 2. In the case of S7-400 you could get the PSU to take slot 1 and 2 and then CPU is in slot 3. For S7-1200/S7-1500 the CPU is actually slot 1 as there is no actual bus system. For the S7-200 the slot is always 2.

Type – (Default = PG) Options are PG(1) programming interface, OP(2) operator panel , S7(3) standard Scada communications.

4.1.5. MPI/PPI Configuration

Local Address – Local MPI/PPI address, usually 0.

PLC Address – MPI/PPI address of the PLC.

Speed – The speed at which the MPI/PPI connection has been configured.

4.1.6. Test Device Configuration for Connectivity

This is used to check whether or not the configuration currently on the dialog can establish a connection and communicate correctly with the PLC. Use this feature instead of having to start the Agent Server only to find your configuration does not establish a connection or communicate correctly with the PLC.

Test Address – The Siemens PLC address which the driver will attempt to scan.

IO Type – The Adroit type in which returned values from the PLC will be displayed in.

Start Reading – If this button is clicked, the driver will attempt to connect to the PLC using the current configuration on the dialog and the driver datascope will automatically open. If the driver does not connect or does not receive any information back from the PLC within 5 seconds, the scan attempt will be deemed unsuccessful and an appropriate error message will be displayed in the message box to the right of the button.



INFORMATION

Note, be sure to read messages on the device's datascope and read message box on the right-hand side of 'Test Device Configuration for Connectivity'. These messages will help you determine whether the device is operating correctly.

4.1.7. Watchdogs

Enable PLC program running check – Enable to activate.

This is a master side check that will force all I/O offline if the SPU is stopped and not the CP card. Data will become latched and stale. Use this to check for activity.

Address DB:DBW any DBx:DBWy that has activity, if the value is stale in this register for more than 1 second, all I/O on this device will be marked bad.

Enable driver send watchdog – Enable to activate.

The device will periodically send a state to the PLC and is meant for a slave/PLC side check to monitor connections, to reset ports etc.

This watchdog does not conflict with the CPU program check watchdog.

Bitaddress - Use any bit address type, e.g., MBX100.0, DB0:DBW0.0, etc.

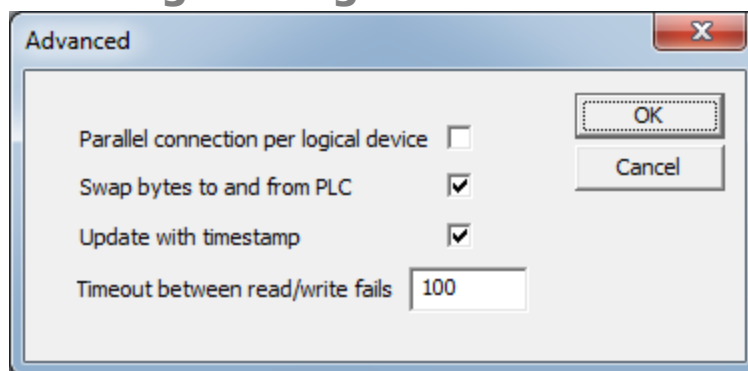
Timeout (mSec) - This is the cycle period in milliseconds to send the next watchdog bit.

Type (options : LatchON/OFF, PulseON/OFF)

The latch options will send the require value continuously once per timeout cycle.

The pulse options will toggle the bit value once per cycle, so a duty cycle of 50% can be assumed.

4.2. Advanced Settings Dialog



Parallel connection per logical device – (Default = unchecked) Only available when (S7-300 or S7-400) and ISOTCP selected. If two or more logical devices (devices configured in Adroit) have been configured to communicate to the same PLC, they will share one connection. If this checkbox is checked, each logical device will create a new connection to the PLC.

Swap bytes to and from PLC – By checking or unchecking this checkbox, the endianness between the driver and PLC is reversed.



PLEASE NOTE

If the values being read from the PLC are vastly incorrect, try checking or unchecking this checkbox.

Update with timestamp – By checking this checkbox, the values in the Agent Server will be updated on every read. With the checkbox unchecked, the values in the Agent Server will only be updated when they change.

Timeout between read/write fails – On the event of a read/write fail, the driver will wait this time (in milliseconds) before trying to read/write again.

5. Supported Addresses

Siemens S5 and S7 PLC data address types supported:

For Data Blocks:

Min Range	Max Range	PLC Data Type	Bool	Int	Real	Str
DB0.DBX 0.0	DB 65534.DBX 65534.7	Digital	X			
DB0.DBB 0	DB 65534.DBB 65534	Unsigned 8		X	X	
DB0.DBW 0	DB 65534.DBW 65534	Unsigned 16		X	X	
DB0.DBI 0	DB 65534.DBI 65534	Signed 16		X	X	
DB0.DBL 0	DB 65534.DBL 65534	Signed 32		X	X	
DB0.DBD 0	DB 65534.DBD 65534	Unsigned 32		X	X	
DB0.DBF 0	DB 65534.DBF 65534	IEEE 32			X	
DB0.DBS 0 S2	DB 65534.DBS 65534 S79	Character String				X
DB0.DBB 0 S2	DB 65534.DBB 65534 S79	Pascal String				X

For Markers:

Min Range	Max Range	PLC Data Type	Bool	Int	Real	Str
MX 0.0	MX 65534.7	Digital	X			
MB 0	MB 65534	Unsigned 8		X	X	
MW 0	MW 65534	Unsigned 16		X	X	
MI 0	MI 65534	Signed 16		X	X	
ML 0	ML 65534	Signed 32		X	X	
MD 0	MD 65534	Unsigned 32		X	X	
MF 0	MF 65534	IEEE 32			X	
MS 0 S2	MS 65534 S79	Character String				X
MB 0 S2	MB 65534 S79	Pascal String				X

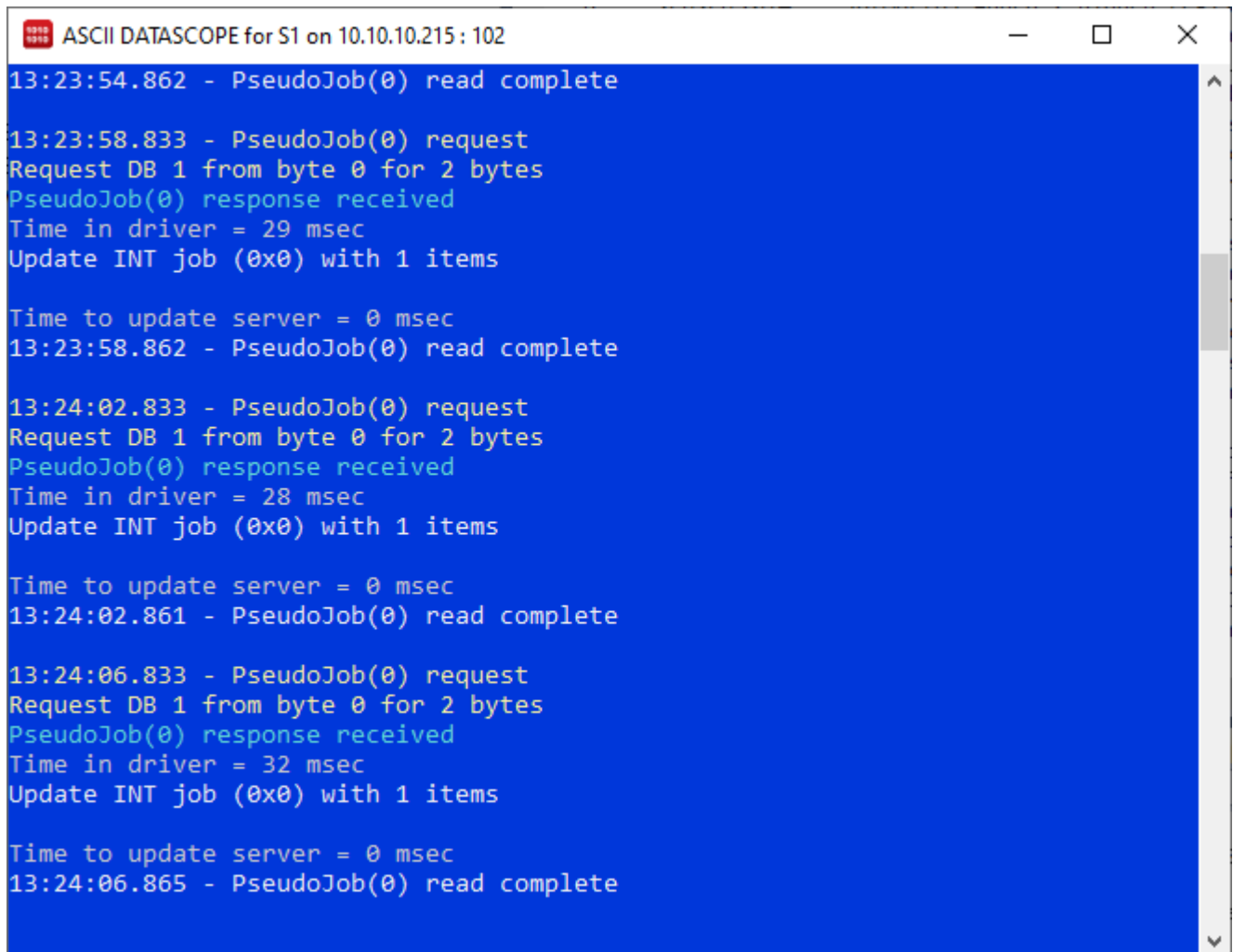
For Inputs/Outputs:

Min Range	Max Range	PLC Data Type	Bool	Int	Real	Str
I/QX 0.0	I/QX 65534.7	Digital	X			
I/QB 0	I/QB 65534	Unsigned 8		X	X	
I/QW 0	I/QW 65534	Unsigned 16		X	X	
I/QI 0	I/QI 65534	Signed 16		X	X	
I/QL 0	I/QL 65534	Signed 32		X	X	
I/QD 0	I/QD 65534	Unsigned 32		X	X	
I/QF 0	I/QF 65534	IEEE 32			X	
I/QS 0 S2	I/QS 65534 S79	Character String				X
I/QB 0 S2	I/QB 65534 S79	Pascal String				X

A character string reads character for character and return it as a string. Pascal string is the standard Siemens string which has 2 bytes in front of the actual string data bytes which contains the character 'P' as first byte and the string length in the second byte.

6. Protocol Definition and Considerations

6.1. Datascope



```
ASCII DATASCOPE for S1 on 10.10.10.215: 102

13:23:54.862 - PseudoJob(0) read complete

13:23:58.833 - PseudoJob(0) request
Request DB 1 from byte 0 for 2 bytes
PseudoJob(0) response received
Time in driver = 29 msec
Update INT job (0x0) with 1 items

Time to update server = 0 msec
13:23:58.862 - PseudoJob(0) read complete

13:24:02.833 - PseudoJob(0) request
Request DB 1 from byte 0 for 2 bytes
PseudoJob(0) response received
Time in driver = 28 msec
Update INT job (0x0) with 1 items

Time to update server = 0 msec
13:24:02.861 - PseudoJob(0) read complete

13:24:06.833 - PseudoJob(0) request
Request DB 1 from byte 0 for 2 bytes
PseudoJob(0) response received
Time in driver = 32 msec
Update INT job (0x0) with 1 items

Time to update server = 0 msec
13:24:06.865 - PseudoJob(0) read complete
```

7. General Notes and Troubleshooting

7.1. Contact Details

Email: support@adroit.co.za and drivers@adroit.co.za

Phone: +27 11 658-8100

Web: www.adroit.co.za

7.2. Adroit Technologies Knowledge Base

For additional information, please refer to our Knowledgebase website:

[Latest Drivers topics - Features, discussions, tips, tricks, questions, problems and feedback \(adroittechnologiesautomation.com\)](http://adroittechnologiesautomation.com)

7.3. Troubleshooting

7.3.1. Error 33028 – cannot connect to the Siemens S7-1200/1500 PLC

This error indicates the following: "The S7-1200 has indirect access on by default. Check if Optimised Block access is disabled." To resolve this issue, ensure that all the steps in Appendix D - Creating S7-1200/1500 Ethernet Connection have been correctly implemented.

8. Appendix A – PLC Models Tested

The following PLC CPU's and cards have been tested so far:

S7-200 CPU 226 with CP Card CP 243 – 1.

S7-300 CPU 314C-2 with CP Card CP 343 – lean.

S7-400 CPU xxx

S7-1200 and S7-1500 (WARNING – See [Appendix D](#) below for correct setting)

9. Appendix B – Considerations

The Siemens driver is capable of reading from multiple data blocks, markers, inputs and outputs in a single transaction. This capability greatly reduces the number of transactions needed to fetch data scanned in Adroit from a PLC.

To support this multi access read capability, the Siemens driver has an internal polling thread. This thread is responsible for polling the PLC. Note, Adroit usually manages polling (not the drivers). In order to fill each PDU to maximum capacity, scan jobs which have whose 'wait' period have elapsed, are dynamically filled into the PDUs.

This architecture can result in scan jobs which should be scanned in the same read, not being scanned at the same time. This circumstance can only occur during online adding of scans. To rectify this, if online scans are added, simply stop the Adroit device, wait the maximum scan time, then start device again. The driver will have optimized which jobs should be scanned together.

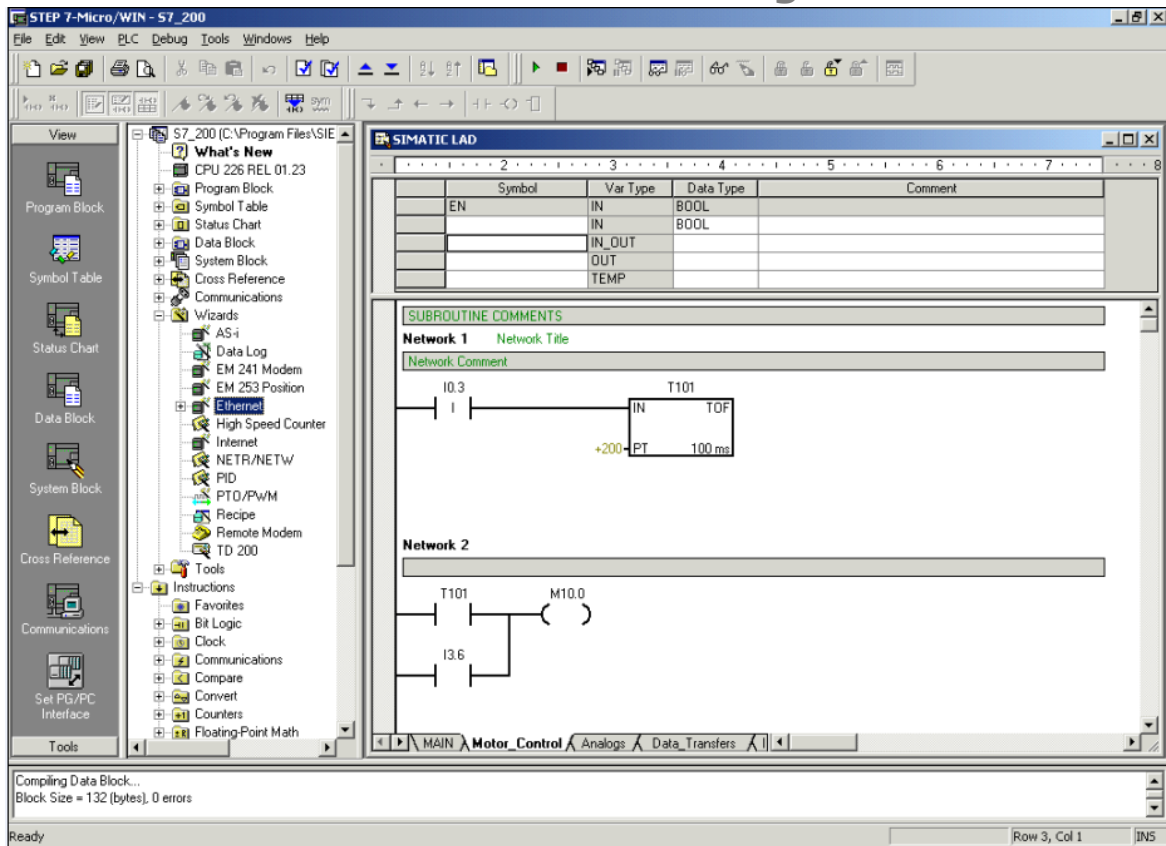


PLEASE NOTE

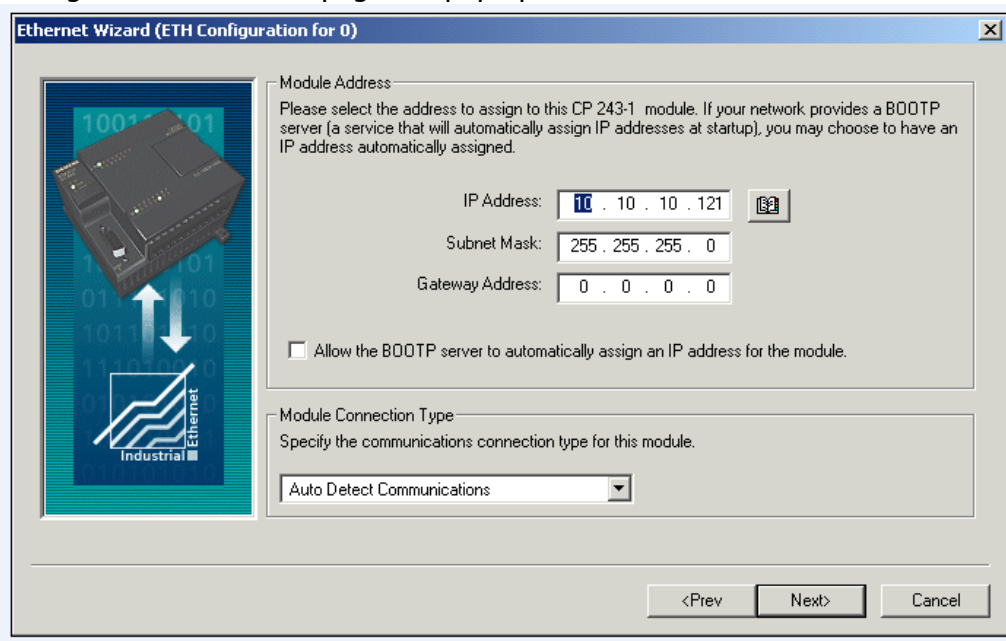
Please note, this can ONLY occur if online scans are added. When the Adroit Server is started, the driver automatically optimizes which scan jobs should be read together.

10. Appendix C – Creating S7-200 Ethernet Connection

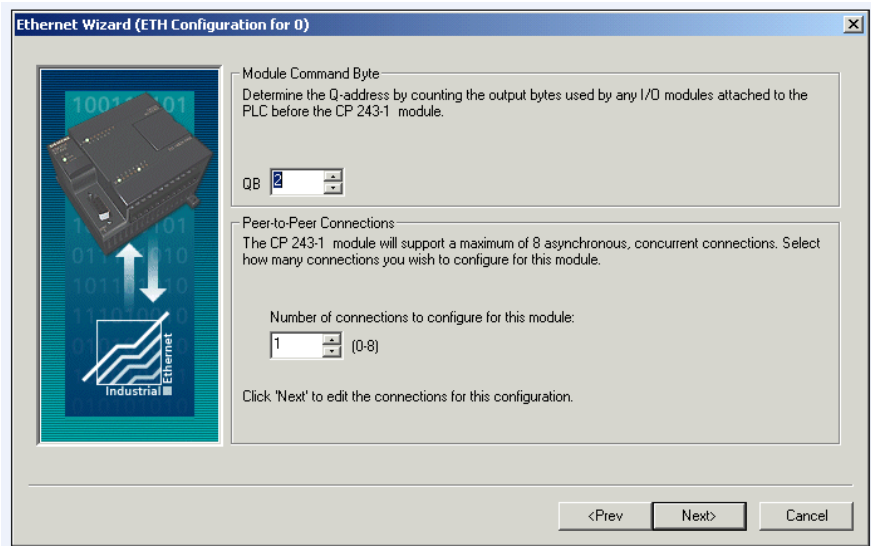
10.1. Create a new Ethernet Configuration



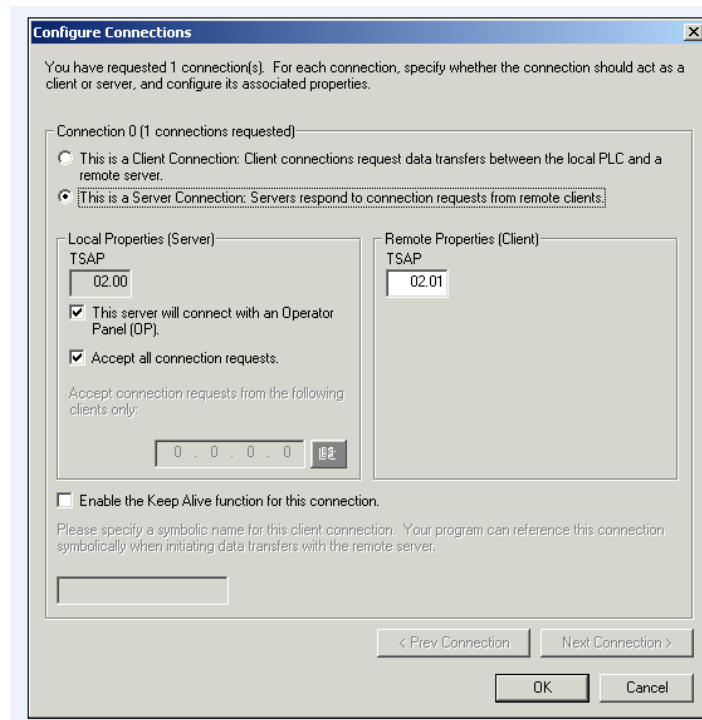
When using MicroWin 4, right click with the mouse on the Ethernet Wizard as shown above. Select 'New Configuration'. The next page will pop up.



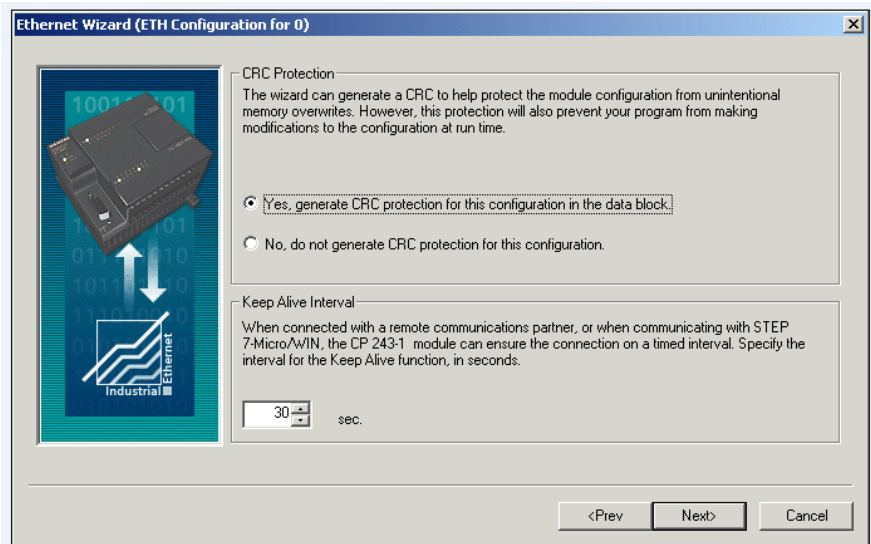
Specify the Ethernet module IP Address and then press next. The next page will pop up.



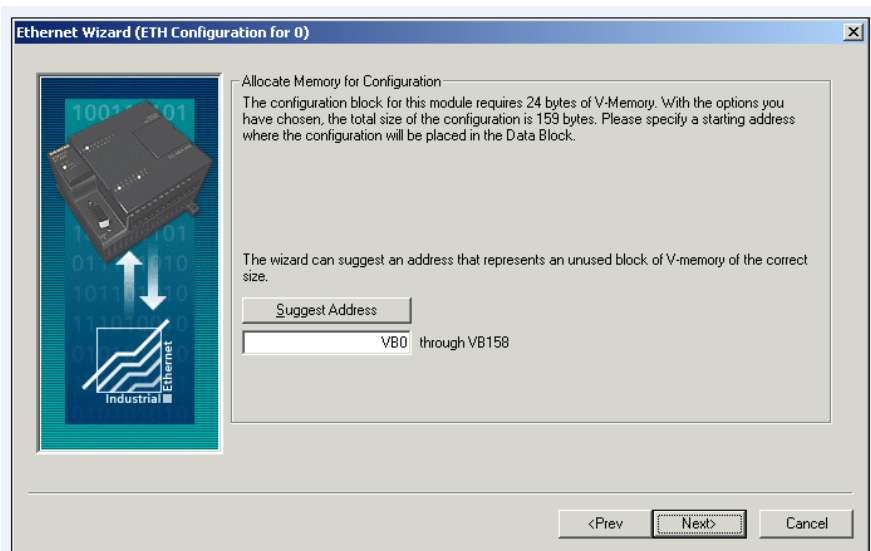
Specify the number of connections required. At least 1 for Adroit connection. Now press next button to open the next screen.



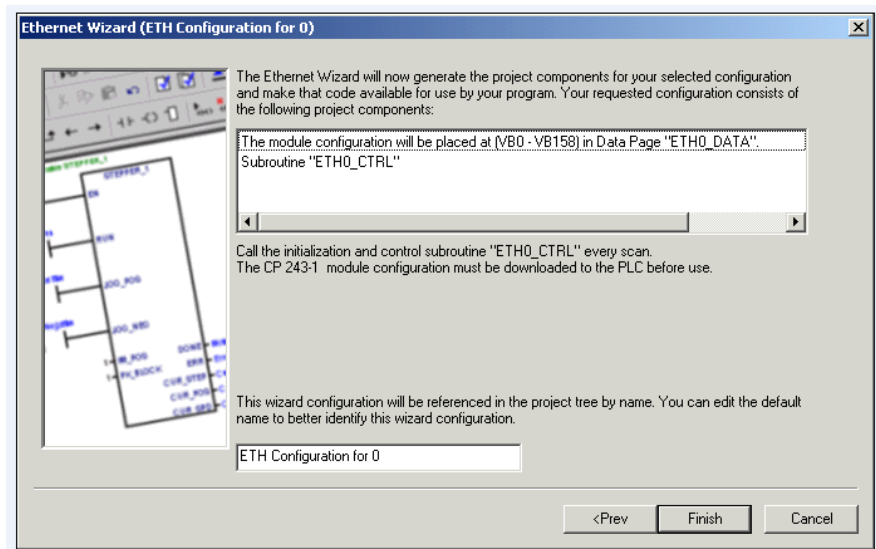
This is the most important screen of all. This is where we setup the connection to allow Adroit to communicate to the PLC. The screen above is the exact setting required to make the communications work. We want a server connection that allows the PLC to operate as the server and Adroit as the client which connects and requests data from the PLC. The Local TSAP on this page refers to the PLC TSAP settings. This setting is not changeable and when 'The server will connect with an Operator Pane(OP)' is selected the TSAP will change to 02.00. This means the SLOT setting in the driver is 2 and RACK setting is 0. Also make sure 'Accept all connection requests' is checked. This will allow for any IP address to create a connection to the PLC. The Remote TSAP properties refer to the PC TSAP and always leave it as 02.01. Once ready click the OK button to continue to the next screen.



Generate CRC protection, it is a safer option. Click next button to continue to the next screen.



This will just allocate memory area for the configuration, leave as default. Click next button to continue to the next screen.



This is the last screen. Here you can give a name for your configuration if so do required but leave it as default otherwise. Press finish button to complete the new Ethernet connection setup. After this setup is completed make sure to press the 'save all' button. Now go to the PLC menu and press 'Compile All' item. This will compile and check everything. If there are no errors, press the download button to download to the PLC. Adroit should be able to communicate to the PLC.

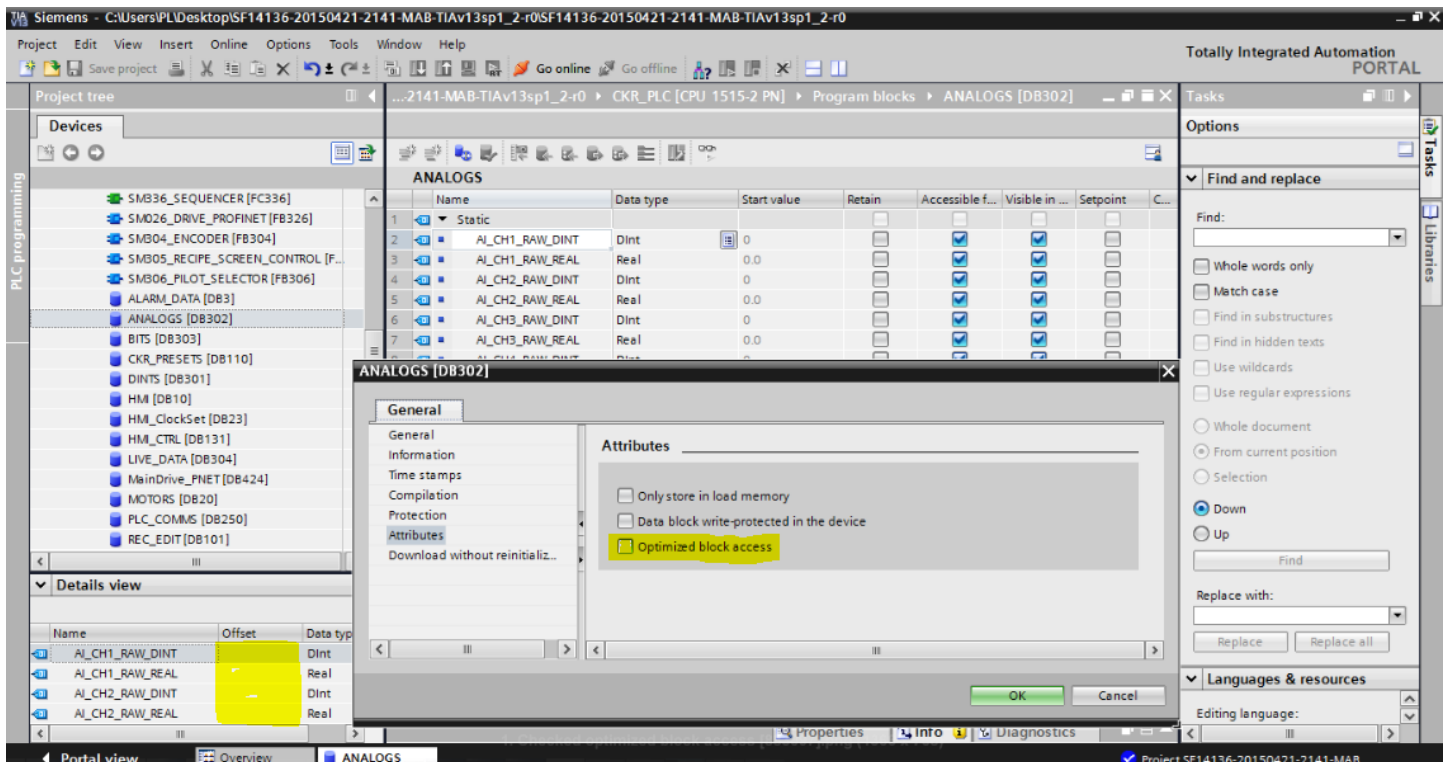
11. Appendix D – Creating S7-1200/1500 Ethernet Connection

11.1. Configuring Siemens TIA Portal Software

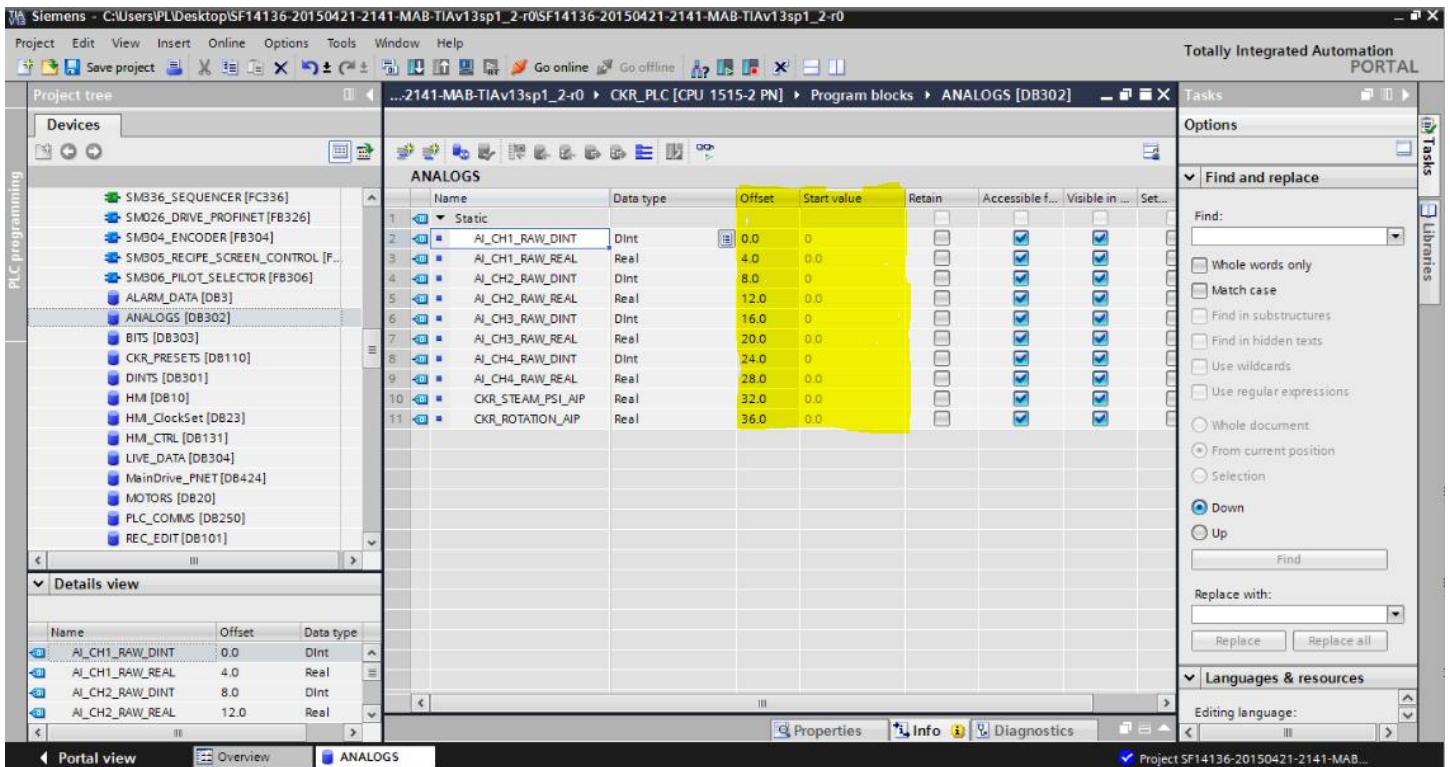
See below there are 3 points to consider: **A, B, C**

From a Scanning Address point of view it is important to be able to see the offsets in a DB.

In the new TIA software this is possible only if you uncheck the DB Setting called **Step A: "Optimized block access"**. By default, it is checked so once it is unchecked you will see the Offset in the left bottom corner as per the image.



Once the Offset is displayed you can compile the block and be able to also see the start value of the variables inside the DB.



The most important setting for the driver to communicate is in the CPU Properties. Select the "Protection" option under the General TAB. Ensure **Step B "Full access (no protection)"** selection box is checked as well as the **Step C "Permit access with PUT/GET communication from remote partner (PLC, HMI, OPC, ...)"** checkbox is checked as per the image below.

